9/8/23

Book keeping → unstructured, may not be audited

Accounting → follow guidelines, principal based book keeping

Financial statements → can be audited

		A	B
1.	Revenue	₹1000 crore	₹1000 crore
2.	Loan	₹100 crore	₹10 crore
3.	Total Assets	₹2000 crore	₹2000 crore

Existing shareholder	External / Finance by Institution
a) Shift from A → B	
b) Dividend (PAT %)	
↳ from PAT	

Revenue - Expenses = Gross Profit

Gross profit - Interest - Tax = Profit After Tax (PAT)

Balance sheet
P&L A] static measure

Cash flow] dynamic measure

10/8/23

Financial Statement

- Balance sheet

Principles of balance sheet

- Entity - for companies only, not for sole enterprises
- going concern
- Accrual basis - Ex. if company has written a cheque & it takes 2 days to clear

↓

accounting follows conservative approach, whenever something happens you record it

cheque

50k

T=0

Cash (T=0) ₹2,00,000

(-50,000) = ₹1,50,000 → In balance sheet,

Bank (T=0) "

in actual bank account it will be reflected after 2 days

- Dual Aspect

Total assets = Total liabilities + owners equity

Assets - liabilities = Owners Equity

- Consistency of report

Ind AS → Indian accounting standard
US AS
GAAP → Generally accepted accounting principles

Assets → something from which you can derive future benefits both monetary & non monetary

- Land, property & equipment - NCA
- Capital / cash - CA
- Patents, goodwill - NCA → will take time
  ↳ something for society
- Inventory - CA
- Subsidies - NCA
- Prepaid expenses - CA
- Account recievables - whenever items are sold on credit basis
  - CA
  ↓
  but why?
  → to increase their sales
- Investments - CA (can liquidate at any point of time)
  (securities)
  ↓
  want to create portfolio for income

Current assets (CA)

Non current assets (NCA)

Inventory — Raw material
— WIP (Work in progress)
— FG (Finished goods)

↓

already ready,
problem may be with demand & supply, not with the product
so that's why CA

Boeing
$\boxed{\text{Aeroplane}}$ — Leasing
|
$\boxed{\text{Indigo}}$

11/8/23

Liabilities - obligations (always see wrt obligations)

1. Accounts payable - CL
2. Loans (Term loan / Long term borrowings) - NCL
3. Wages / Salary - CL
4. Deferred tax liability - company has recognised that it will need to pay taxes in future
   → tax obligation is accumulated, but is due in subsequent
   NCL
5. Current / short term loan - CL
6. Owners equity - Equity capital
   ↳ how much money has been given by shareholder / promoter

Assets	\$ (mn)
- Current Assets	
a) Cash	1,449
b) Marketable securities	246
c) Inventories	9,944
d) Pre-paid expenses	10,623
e) Account Recievables	389
	22,651

Marketable securities are financial assets that can be easily bought and sold on a public market, such as stocks, bonds, and mutual funds. These securities are listed as assets on a company's balance sheet because they can be easily converted into cash.

- Non current assets

a) Property, plant, Equipment (PPE) 26,946

- depreciation 13,534

PPE, net 13,412

b) Long term investment 1,110

c) Patent & trademark 403

d) goodwill 663

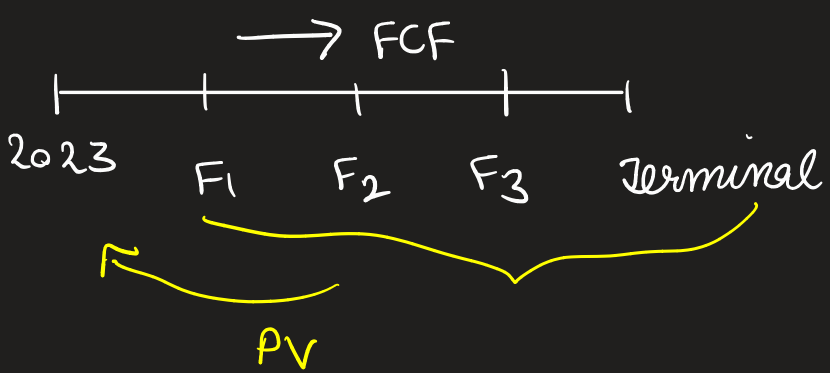
NCA

Total assets ← 38,239

A_A B_T → \$250 mn fair value

\$ 300 mn actual value

Actual value - fair value = goodwill



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Treasury bills → Issued by central bank on behalf of gov.
- By government to raise capital
- 91, 182, 364 days durations

Imp

Shareholder's equity → liability → Why?

↳ bcz when shareholders give money, its kind of a loan

Retained earning → money / profits left after giving to shareholders

Income statement → Profit & loss statement

Net income / profit = Profit - Tax

↓

belongs to existing shareholders

Retained earning → every year retained earning is cummulative after making adjustments for dividends

Gross margin = Net sale - cost of goods sold (cost of sales)

Net sale = Gross sale - returns - sales discounts

Total sales

↓

Total revenue ↓ come down like this

⋮

↓

Net profit

SEC → In USA 3 years for income statement & 2 years for balance sheet

In India → 2 years → for both income statement & balance sheet

\# of O/S ⇒ 2 lacs

PAT → Profit after tax

₹20 lakhs

Basic EPS = $\frac{PAT}{\# O/S} = \frac{20 \text{ lacs}}{2 \text{ lacs}} = ₹10$

↳ shares which you can see at any point of time

EPS → Earnings per share

Diluted EPS = $\underbrace{2 \text{ lacs} + 1 \text{ lacs}}_{3 \text{ lacs}}$ (warrent / convertibles)

↓

instruments which can be converted into shares

$= \frac{20 \text{ lacs}}{3 \text{ lacs}} \simeq ₹7$

17/8/23

- Balance sheet
- Income statement (P&L statement)

Cash flow

Cash

Source

→ Operation

→ Investment

→ Financing

Uses

Net Cash

$$\text{Gross margin / Gross profit} = \frac{\text{Sales} - \text{COGS}}{\text{Sales}}$$

$$\text{Profit margin} = \frac{\text{Net income}}{\text{Revenue}} \times 100\%$$

Sales

− (COGS)

Gross profit

(EBITDA)

↳ Earnings before interest, taxes, depreciation, amortization

1/9/23

Financial Statement Analysis

Earnings price ratio $= \dfrac{EPS}{MP}$

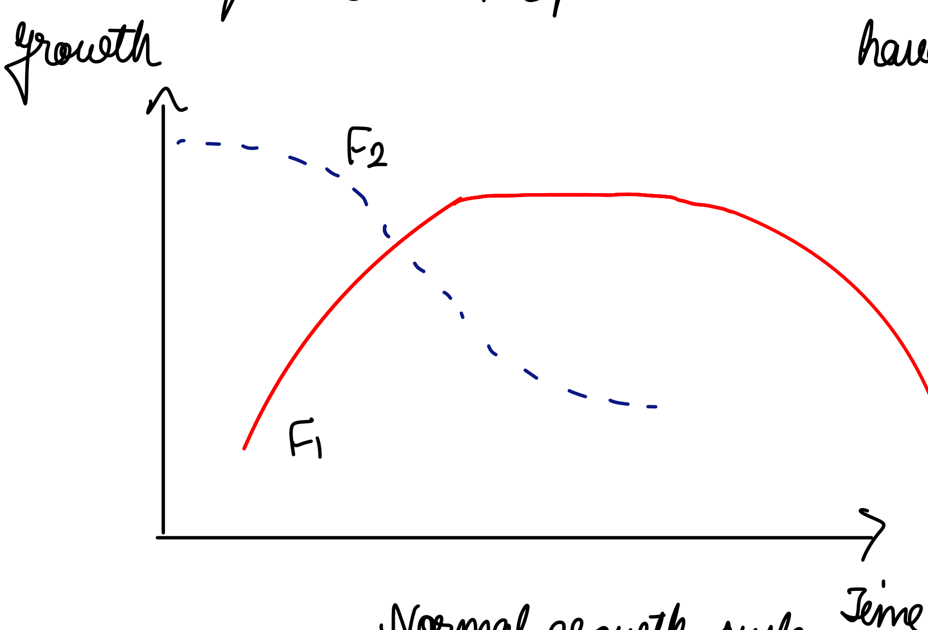
$$\boxed{PE\ ratio = \frac{MP}{EPS}} \rightarrow \text{Market price}$$

2010	1997	
F_1	F_2	I_{PE} → Industry PE ratio
3	10	7

→ For every ₹1 earning, you are paying ₹10

F_2 ⇒ overvalued, so low chance it will go up, so should sell the stock and get F_1

If no information assymetry is there, no ambiguity then whatever earning company has generated should be equal to market price → PE would be 1, you won't have to pay high value



Normal growth cycle

PE ratio of F_1, F_2 is different, but both F_1, F_2 might be at different stages of their growth

So, we use

$$\boxed{PEG = \frac{PE}{growth}}$$

Dividend payout ratio → how much dividend company is paying out of profit

$$\text{Dividend payout ratio} = \frac{\text{Dividend} \times 100}{\text{Net income}}$$

$$\boxed{DPO = \frac{DPS}{EPS}}$$

$$= \frac{(\text{Dividend}/N)}{(\text{Net Income}/N)}$$

Turnover ratio → efficiency in producing goods

Working capital turnover → how short term assets are used to produced finished goods

Shareholder's equity turnover ↑, better efficiency

Solvency ratio
- Short term
- Long term

Current ratio : $\dfrac{\text{Current assets}}{\text{current liabilities}}$

↓

should be greater than 1

Quick ratio → Remove inventory, raw materials from current assets

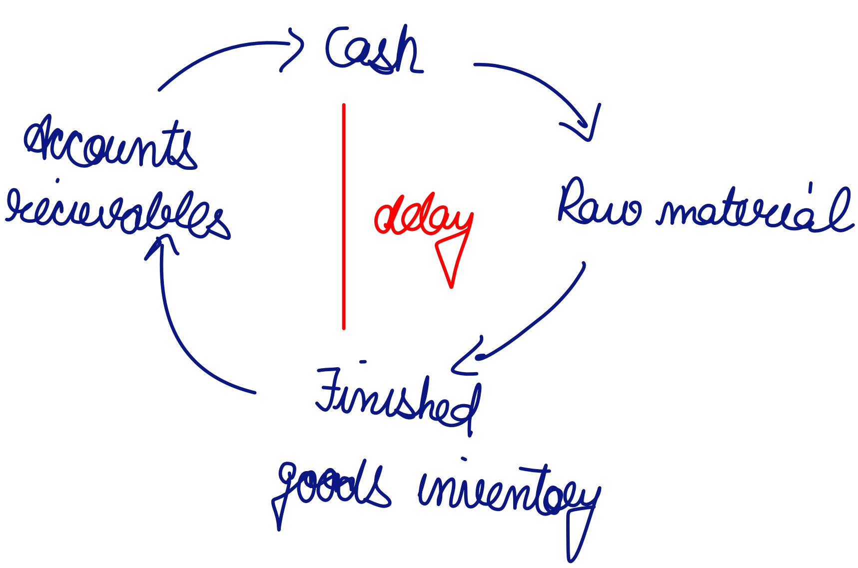
↳ so gives ratio of highly liquid current assets to current liabilities

Q Vinyl chemicals problem

Profitability ratios: RoI, RoE, RoTA, EPS, DPS

Solvency (short term): NWC, CR, QR

There is a delay is getting cash and finishing goods



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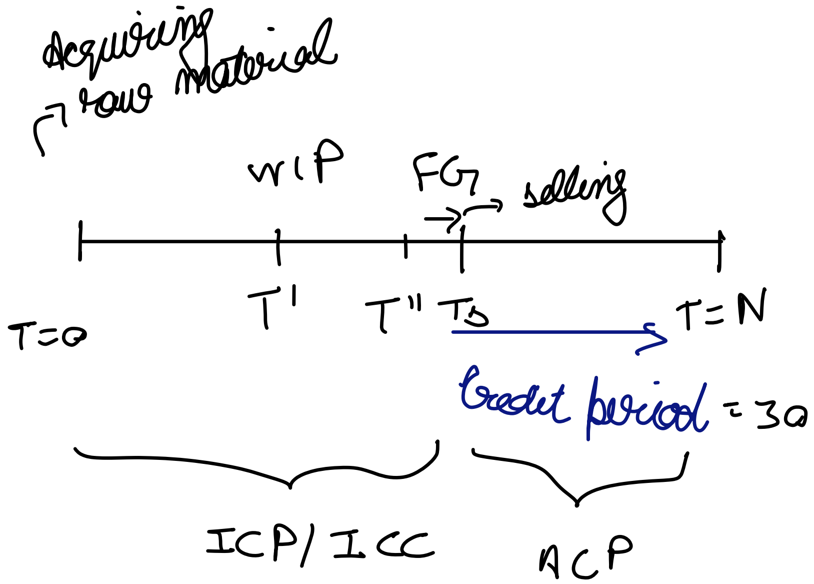
Financial Statement Analysis



T_s = selling time
= 15

ICP → Inventory conversion period

ACP → average collection period

APP → Average payable period

ideally should be short so that can recieve more cash in short period ← OC − APP = cash conversion cycle

APP → Average payable period

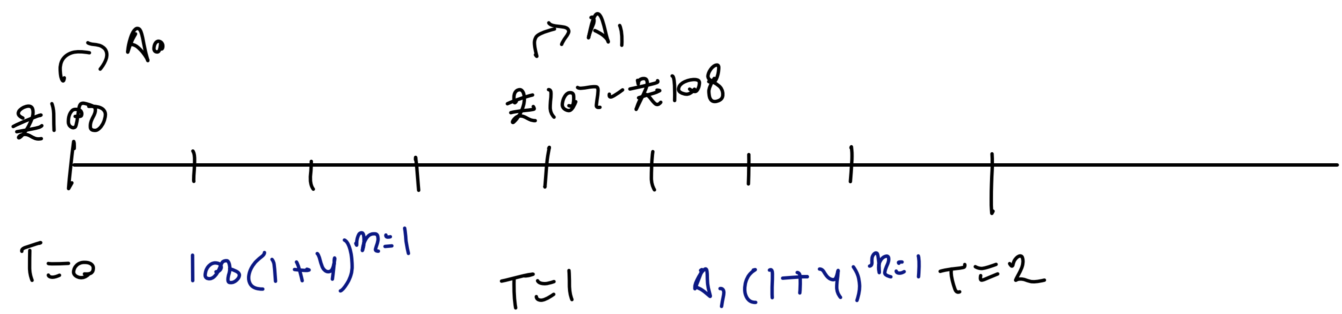
OC = ICP + ACP

$$ROE = \boxed{\frac{\text{Net income}}{\text{Sales}}} \times \boxed{\frac{\text{Sales}}{\text{Total assets}}} \times \boxed{\frac{\text{Total assets}}{\text{Net worth}}}$$

operational efficiency — utilization of assets — financial leverage (how creating value for shareholders)

Du Pont analysis → breaks into 3 parts

Time Value of Money



$A_0(1+y)^{n=2}$

① Compounding
- Lumpsump → one time cash inflow
- Annuity →
- Multicompounding

Nominal interest rate = Real interest rate + Inflation

Bank

↳ Quaterly

Interest rate = 8%

$$EIR = \left(1 + \frac{y}{\text{freq}}\right)^{\text{no. of compounding}} - 1$$

$$EIR = \left(1+\frac{0.08}{4}\right)^4 - 1$$

$$= (1.02)^4 - 1$$

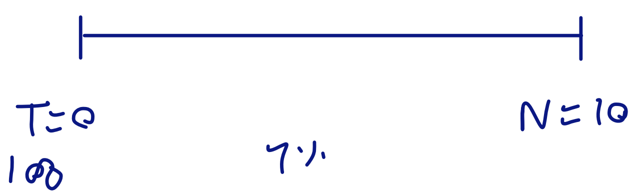
$$= 0.0824$$

$EIR = 8.24\%$

↳ the extra interest we are getting is due to compounding

① lumpsum

↳

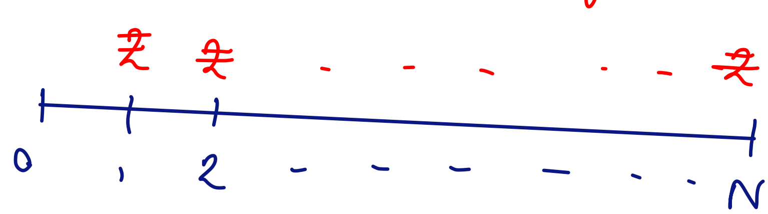


→ Future value $= 100(1+0.07)^{10}$

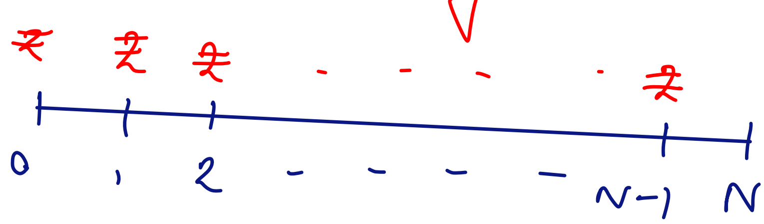
② Annuity - at the end of every time period, we get a fixed amount

- Annuity
- Annuity due

→ only difference is annuity due starts from t=0 till N-1 whereas annuity starts from t=1 to N



Annuity



Annuity due

Net income ₹20 million   P/E -> 8.2

EPS = 10

dividend : ₹5.74

Dividend payout -> from profit

how much dividend is paid

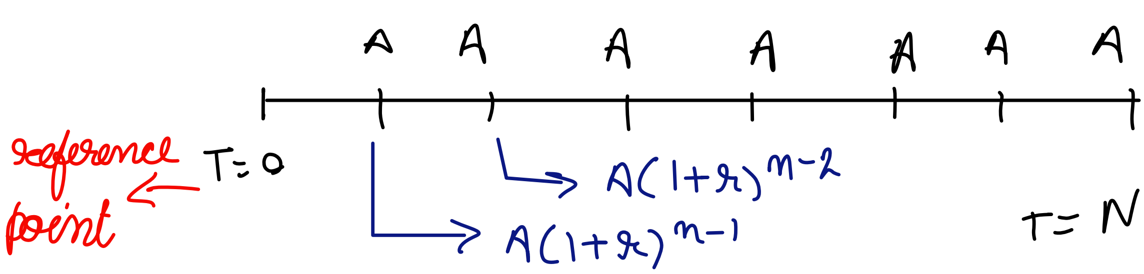
13/9/23

- FV of lumpsump
- PV of lumpsump
- effect of multicompounding

$$FV = A \times \underbrace{(1+r)^n}_{\text{Future value interest factor (FVIF)}}$$

$$PV = A / \underbrace{(1+r)^n}_{\text{PVIF (Present value interest factor)}}$$

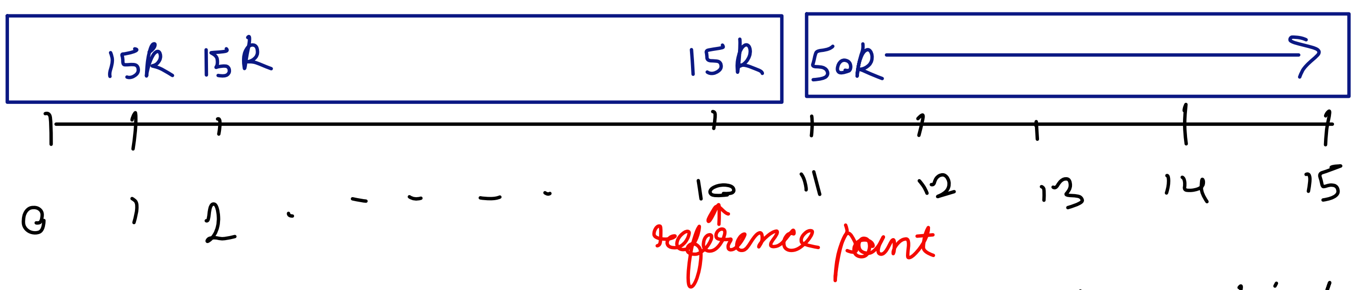
$$EIR = \left[\left(1+\frac{r}{m}\right)^m - 1\right]$$



$$FV = A \underbrace{\left[\frac{(1+r)^n - 1}{r}\right]}_{\text{FVAF (Future value annuity factor)}} \rightarrow \text{sum of all cash flows}$$

$$PVA = A\underbrace{\left[\frac{1-(1+r)^{-n}}{r}\right]}_{\text{PVAF (Present value annuity factor)}}$$

If PVA > amount invested → then fine, else no benefit



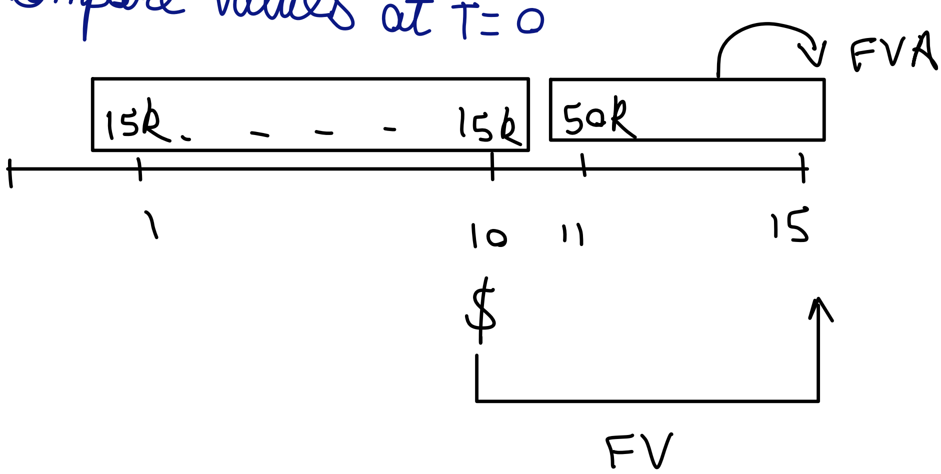
Important thing is how you choose your reference point

(1) $$15k\left[\frac{(1+0.07)^{10}-1}{0.07}\right] = 50k\left[\frac{1-(1+0.07)^{-5}}{0.07}\right]$$

comparing two different cash streams

(2) compare values at T=0

(3)



→ this assumes you get interest even after 10^{th} year

Assuming → you get 7% after 10 year for 11^{th} – 15^{th} year

So, we have three approaches → three reference points
T=0, 10, 15

But T=10, is better as that has least assumptions

$$FVA = A\left[\frac{(1+r/m)^{n\times m} - 1}{r/m}\right]$$

$$PVA = A\left[\frac{1-(1+r/m)^{-n\times m}}{r/m}\right]$$

→ If amount is compounded more than once in a year

$r \rightarrow r/m$

$n \rightarrow n \times m$

Annuity due

$$FVA_{due} = A\left[\frac{(1+r)^{n}-1}{r}\right](1+r)$$

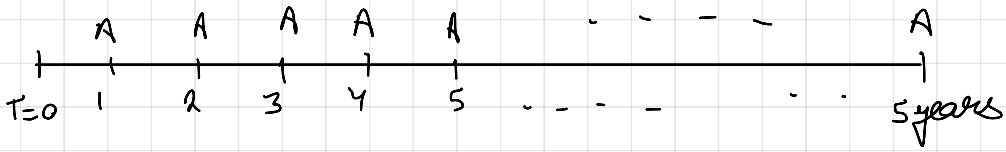
$$PVA_{due} = A\left[\frac{1-(1+r)^{-n}}{r}\right](1+r)$$

15/8/23

$FV = A(1+r)^n \rightarrow$ compounding
$PV = A/(1+r)^n \rightarrow$ discounting

PVIF $\rightarrow \frac{1}{(1+r)^n}$

EMI ⇒ Equated monthly installment → at the end interest + principal has been paid



No. of payments $= 5 \times 12$
$= 60$

Ref $T = 0$

$$10 \text{ lacs} = PV \text{ of EMIs}$$
$$= A\left[\frac{1-(1+r/12)^{-n \times 12}}{(r/12)}\right]$$
$$= A \underbrace{\left[\frac{1-(1.01)^{-60}}{0.01}\right]}_{44.95}$$

$A \simeq 22244$

EMI = principal + interest
Interest = EMI × 60 − loan amount
≃ ₹3.24 lacs

Amortization → a payment we make to write off an asset
- → Maturity
- → Interest
- → Principal amount

Loan of 10 lacs @ 12%

5 lacs @ 10% 5 years

Ammortization payment → $5{,}00{,}000 = A \times \left[\frac{1-(1+0.10)^{-5}}{0.10} \right]$

Amortization schedule → a table consisting all principal + interest payments

Yr	O/S amount	Payment	Principal	Interest	O/S amount END
0	5,00,000				
1	5,00,000	131,000	81,000	50,000	4,19,000
2	4,19,000	"	89,1000	41,900	3,29,900
3	3,29,900		98,010	32990	
4	2,31,890		107,811	23,189	
5	124,079		118592.1	12,407.9	
6	5,486.9				
7					
8					

not coming zero as Payment we have rounded off

Inferences

→ Principal component in payment increasing, as interest is decreasing

→ In balance sheet companies deduct amortization amount from earnings

EBITDA

↳ Earnings before interest, tax, depreciation, amortization

Q

Loan amount	65,00,000
Interest	12%
Years	15

Calculate EMI

Valuation of bonds & stocks

Bond

- → Valuation of coupon bond / zero coupon
  ↳ no intermediate payment
- → YTM (yield to maturity)
- → CY (current yield)
- → Duration
- → yield to call (YTC)

$$B_0 = \underbrace{CP\left[\frac{1-(1+r/m)^{-n\times m}}{r/m}\right]}_{PVIFA} + \frac{MV}{(1+r/m)^{n\times m}}$$

B_1 (AAA)

B_2 (BBB)
↳ must offer higher coupon rate
risk must be compensated by additional coupon

\$1000 − δ ———————— \$1000 → zero coupon

T_0 ———————— T_N

δ → returns

YTM → return generated if held till maturity

Perpetual bond

$$R = \frac{INT}{R}$$

Yankee bond → issued in US, by non US companies for raising capital

Bull dog → In UK

Kimchi

Yield to Maturity vs Current Yield

↓

how much return has been generated in a year

$$= \frac{\text{Annual coupon payment}}{\text{Market price}}$$

Duration of bond

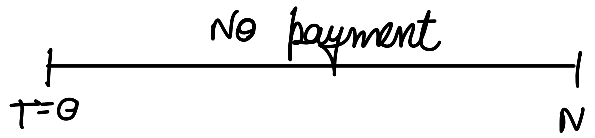
↳ measure of riskiness / sensitivity

How much time it will take to recover intrinsic value / PV of the investment

Portfolio Immunization

Duration of assets = Duration of liabilities

Zero coupon



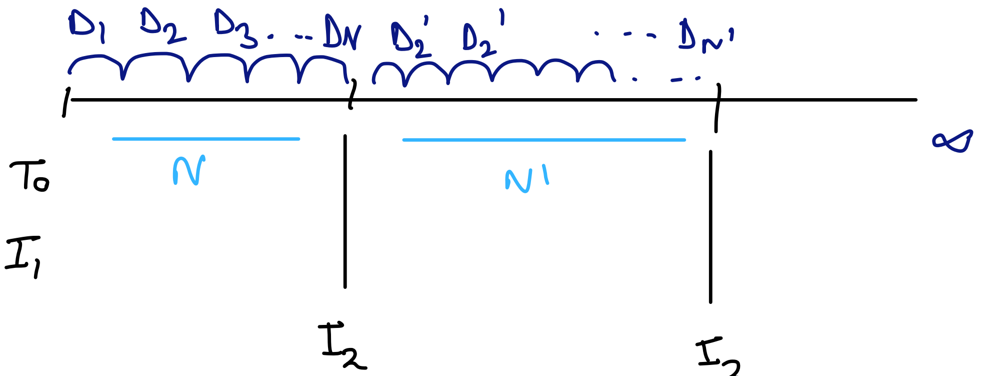
Maculay's duration

Yield curve → It is a kind of sentiment indicator

Stock Valuation

Stock vs Bond

Capital gain – if price of share goes up
Dividend gain → company may pay, but not certain



→ We assume that company will be there till forever (going concern)

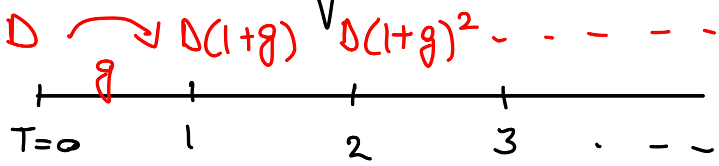
$$P_0 = \sum_{t=1}^{N} \frac{D_t}{(1+r)^t}$$

→ assuming very long time
→ assuming company is paying dividends

→ Constant dividend ①

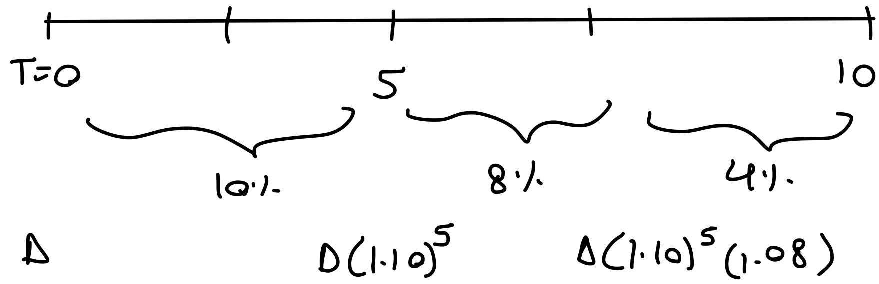
$D_1 = D_2 = D_3 \text{ - - - - } = D$

→ Constant dividend growth ②



g → growth rate

→ Supernormal growth ③



① $P_0 = \frac{D}{r}$

② $P_0 = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g}$

Growth (g) ⇒ b × ROE

b → Retention

ROE → Return on equity

1. Investment → Internally (Retained earnings)

↓

Reinvestment (Retention)

(b)

$\boxed{b = 0.30}$

↳ 30% = 0.3

30% used for reinvestment in projects

2. ROE

$P_0 = \frac{D_1}{r-g} = \frac{D_1}{r - b \times ROE}$ → $r > b \times ROE$ should be

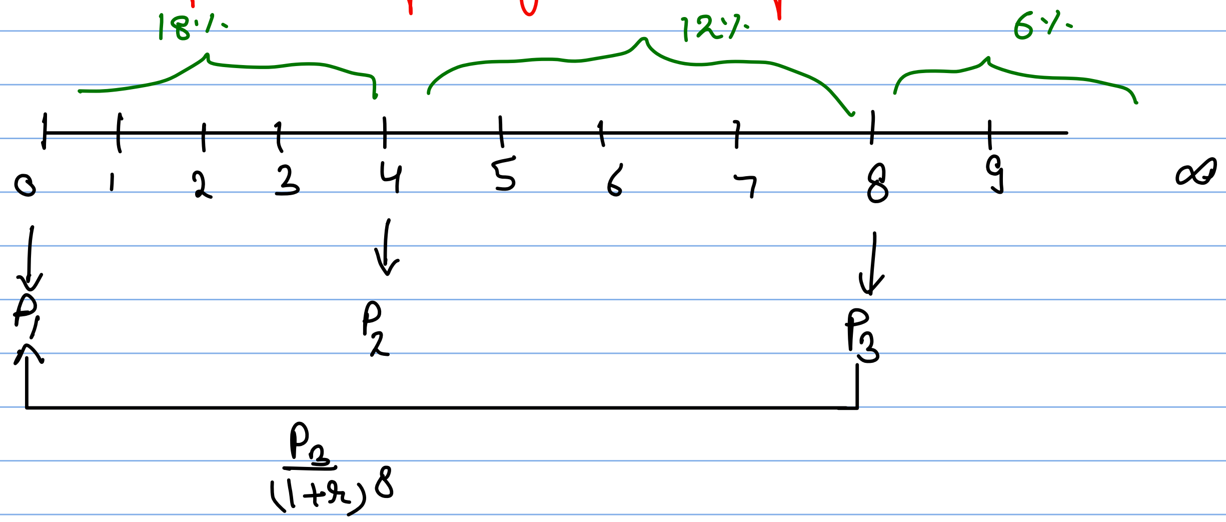
$D_1 = E_1(1-b)$

$$P_0 = \frac{E_1(1-b)}{r - b \times ROE}$$

$$\boxed{\frac{P_0}{E_1} = \frac{1-b}{r - b \times ROE}}$$

6/10/23

current yield → yield for current year



Risk and Return

When we are calculating geometric mean → it considers compounding

SEBI vs SEC
NSE NYSE

1. Spread = fn (information)
   - → Historical trend (price, volume)
   - → Fundamental information (public info)
   - → Insider / Private information

Higher information assymetry → more lubricant (Market Microstructure)

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Portfolio (more than one security)

$A(R_A, \sigma_A)$ $B(R_B, \sigma_B)$

covariance (R_A, R_B)

Return
↑
$$E(R_p) = w_A E(R_A) + w_B E(R_B)$$

$$= \sum w_i E(R_i)$$ → Portfolio return

condition

$w_A + w_B = 1$ $w_i > 0$ → if only 1 security then we can't call it a portfolio

Risk
↑

$$\sigma_p^2 = (w_A \sigma_A)^2 + (w_B \sigma_B)^2 + 2\,cov(A,B)\, w_A w_B$$

$$= (w_A \sigma_A)^2 + (w_B \sigma_B)^2 + 2 C_{AB} \sigma_A \sigma_B w_A w_B$$

Case 1

$C_{AB} = -1$

$$\sigma_p^2 = (w_A \sigma_A)^2 + (w_B \sigma_B)^2 - 2\sigma_A \sigma_B w_A w_B$$

$$\sigma_p^2 = (w_A \sigma_A - w_B \sigma_B)^2$$

$$\boxed{\sigma_p = w_A \sigma_A - w_B \sigma_B}$$

Case 2 $C_{AB} = +1$

$$\sigma_p = (w_A \sigma_A + w_B \sigma_B)$$

→ If market is bearish, your portfolio will go down as both are strongly correlated

Minimum Variance portfolio → assign weights such that risk is minimum

If only two securities in our portfolio

$$\sigma_p^2 = (w_A \sigma_A)^2 + ((1-w_A)\sigma_B)^2 + 2\,cov(A,B)\, w_A (1-w_A)$$

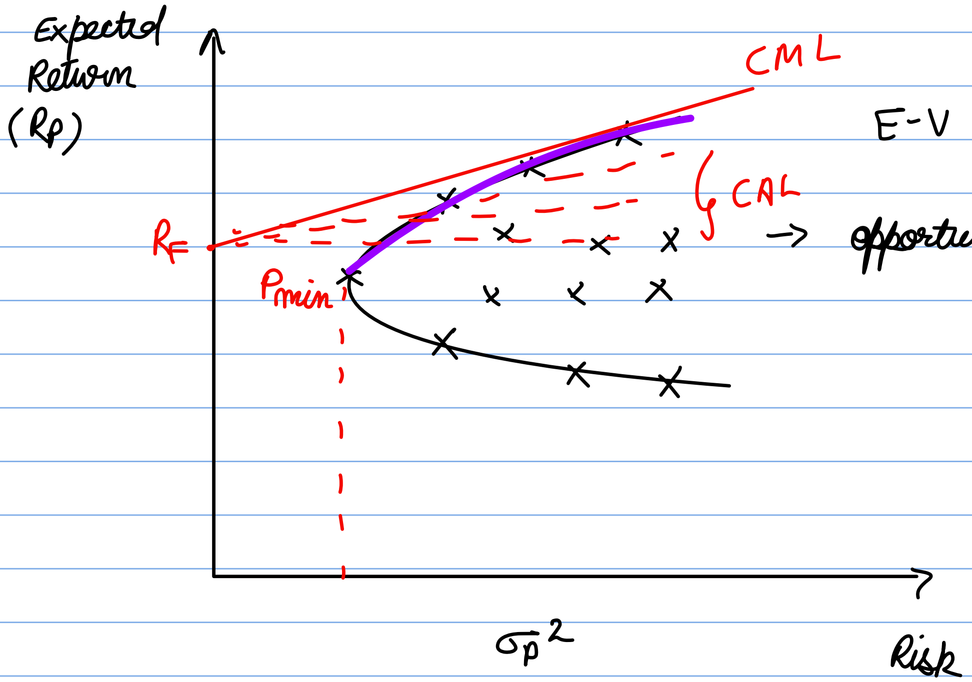
To minimise σ_p^2 we do $\frac{d\sigma_p^2}{dw_A} = 0$

→ Minimum variance portfolio

$$w_A = \frac{\sigma_B^2 - Cov(A,B)}{\sigma_B^2 + \sigma_A^2 - 2Cov(A,B)}$$

$$w_B = 1 - w_A$$

Markowitz (1952) → Portfolio selection



E-V theory

→ opportunity set

Envelope curve

efficient frontier
↓
for same risk you are getting better returns

CAL → capital allocation line

R_F → risk free return
↓
treasury bill

our own portfolio line ← CML → capital market line
↓
CAPM model

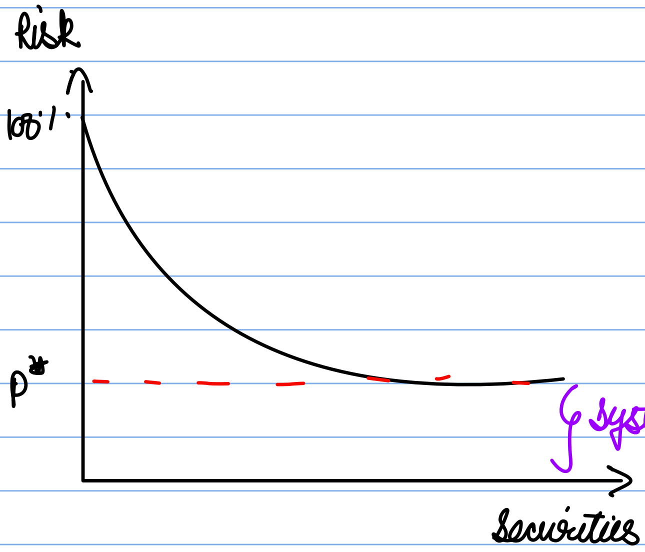
CML → best portfolio given our securities

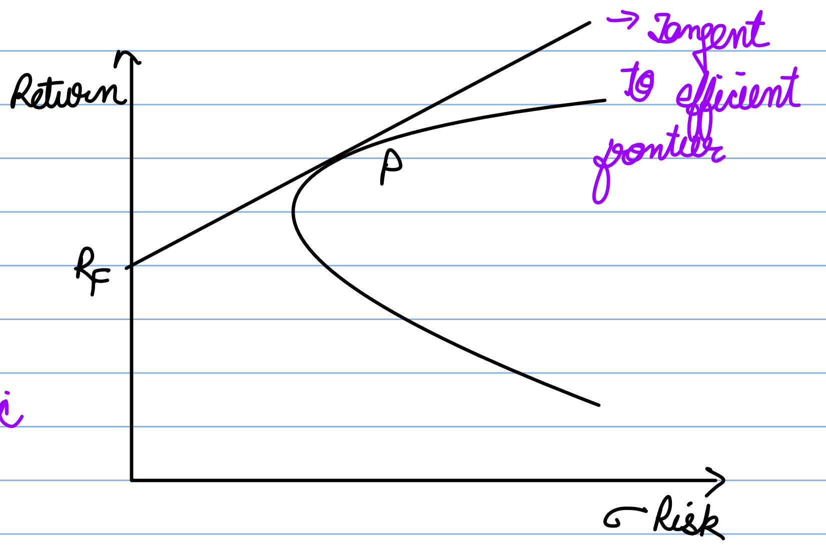
Efficient points – all points above minimum variance (P_{min})

n securities → $n \times n$ variance covariance matrix

12/10

Portfolio Theory - Sharpe





Diversification → we minimise this, can't reduce systematic risk
↳ Unsystematic risk (min.)

$R =$ Certain (Expected return) + Uncertain
↳ unsystematic
↳ systematic

$$\left[\frac{r_i - r_f}{\beta_i} = \frac{r_j - r_f}{\beta_j} = \frac{r_k - r_f}{\beta_k} = \ldots \frac{r_m - r_f}{\beta_m} \right]$$

→ The extra return you are getting for taking risk (extra from risk free return r_f)

β_i ↳ systematic risk

$$\beta_i = \frac{cov(R_i, R_m)}{\sigma_m^2}$$

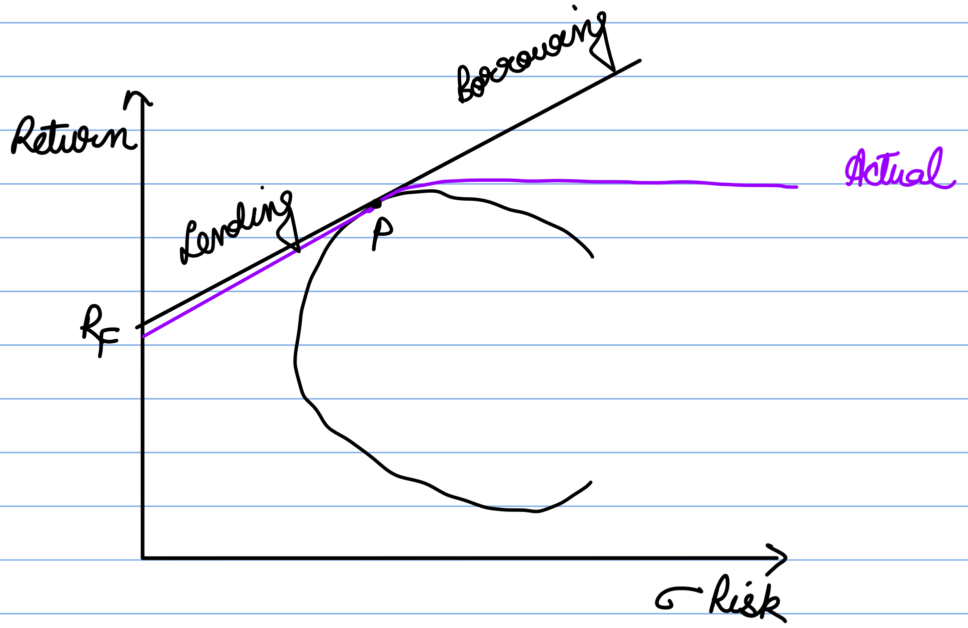
→ Risk premium for diff. securities traded should be same in the same market

$$\beta_m = \frac{cov(R_m, R_m)}{\sigma_m^2} = 1$$

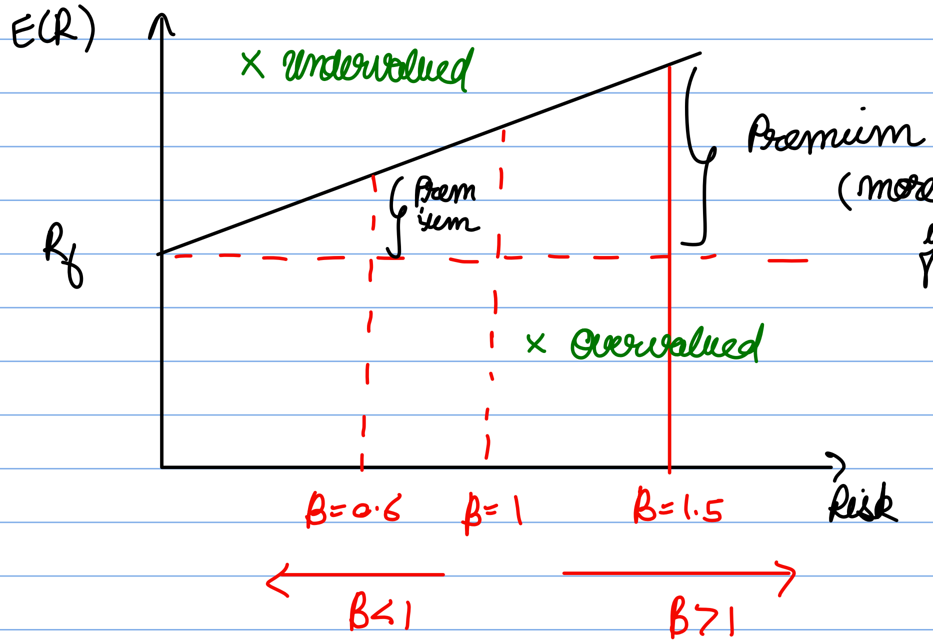
$$\frac{r_i - r_f}{\beta_i} = \frac{r_m - r_f}{\beta_m (=1)} \Rightarrow r_i - r_f = \beta_i (r_m - r_f)$$

$$r_i = r_f + \beta_i (r_m - r_f)$$

β → systematic risk



Premium = Return − Risk free rate



Premium (more than β=0.6, as you take more risk)

Here we have assumed that our stocks lie on capital market line

β<1 → Defensive stocks Ex pharma industry
β>1 → Aggressive stocks

If a stock lies above CML → undervalued
" " " below " → overvalued

In theory → β should not be −ve → as we would get returns less than r_f
↓
But in reality → yes

$\beta_p \rightarrow$ beta of portfolio $\rightarrow$ weighted average of beta's (βs)

When we have to know which investment is better $\rightarrow$ look at slope

CAPM $\rightarrow$ assumes lending and borrowing rate is same $\rightarrow$ but not in real world

→ Hardly 30% stocks are traded actively → in NSE
→ Doesn't account for taxes
→ No brokerage charges

Three type of investors → Risk taking
Risk neutral → least concerned about portfolio
Risk averse

so investors are not homogenous → CAPM assumed homogenous

Modern Portfolio Theory → Elton, gruleen

$$E(R_p) = \sum_{i=1}^{N} P_i R_i$$

$\downarrow$

$E(R_A), E(R_B)$

$$\sigma_p^2 = \sum_{i=1}^{N} P_i (R_i - \bar{R_i})^2$$

$$\sigma^2 = E(X^2) - [E(X)]^2$$

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Capital Budgeting Decisions

- Basic accounting
- TVM
- Valuation
- Portfolio / Return & risk } Investment

- Project for investment
  ↓
  Huge capital is required → so risk is more → so decision making should be based on information
  ↓
  Criteria

EBIT (1 − Tax)

↓

NOPAT / OEAT

gross working capital = Current asset
Net " " = " " − Current liabilities

$$NPV = \sum_{t=1}^{N} \frac{C_t}{(1+r)^t} - C \rightarrow \text{initial investment}$$

$$r = \underbrace{\text{cost of capital}}_{\text{rate at which we have borrowed}} + \underbrace{\delta}_{\text{project specific risk}}$$

NPV → does not consider scale of investment and time also (drawback)

BCR (Benefit cost Ratio) $= \dfrac{\sum_{i=1}^{N} \frac{C_t}{(1+r)^t}}{C_0}$ → Benefit to initial investment ratio

NBCR = BCR − 1
↓
Net Benefit cost ratio

Internal Rate of Return
↳ If IRR > cost of capital → Acceptance

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Discounting Techniques

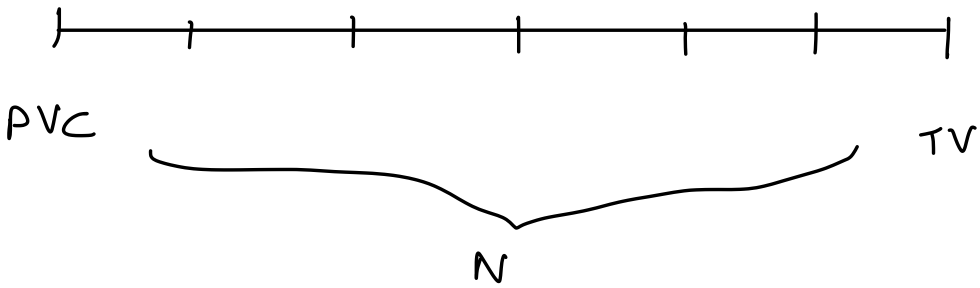
→ NPV
→ BCR or profitability index
→ IRR

NPV = 0
or

$$CRR = \sum_{t=1}^{N} \frac{C_t}{(1+r)^t}$$

→ If there is sign change in cash flows → number of sign changes
"
So what to choose? ← number of different values of IRR you will get
↓
Issue with IRR

In mutually exclusive projects → use Incremental cash flow approach
(A−B)
If +ve → go with A
If −ve → " " B



$$PVC = \frac{TV}{(1+r_{MIRR})^{n}}$$

→ MIRR – Modified internal rate of return

$$PVC = \sum_{t=0}^{n} \frac{\text{Cash outflow}}{(1+r)^{t}}$$

$$TV = \sum_{t=0}^{n} \text{Cash inflow } (1+r)^{n-1}$$

Cost of Capital

1/11/23

2/11/23

Preference share → no voting right

- Irredemable
  - issued for infinite time
- Redeemable
  - may be issued for specific period (maturity time)

WACC → can be estimated using ~~book~~ value, market value ✓
↳ use this

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